Comparison of Data Types in Python

1. Introduction

Data types in Python define the type of value a variable can hold. Choosing the correct data type is important for storing, manipulating, and performing operations on data efficiently. This report compares three basic data types: Integer, String, and Float, with examples and explanations.

2. Integer (int)

Definition:
An integer is a whole number without a decimal point. It can be positive, negative, or zero.

Example in Python:
age = 20
year = -2025
print(age)
print(year)

Explanation:
- Integers are used for counting, indexing, and situations where fractional values are not needed.
- Operations like addition, subtraction, multiplication, and integer division can be performed on integers.

Output:
20
-2025

3. Float (float)

Definition:
A float is a number with a decimal point. It is used to represent fractional or real numbers.

Example in Python:
price = 99.99
temperature = -12.5
print(price)
print(temperature)

Explanation:
- Floats are used when precision is required, like in measurements, prices, or scientific

calculations.
- Operations like addition, subtraction, multiplication, and division can be performed.

Output:
99.99
-12.5

4. String (str)

Definition:
A string is a sequence of characters enclosed in quotes (single ' or double "). Strings are used to represent text.

Example in Python:
name = "Bangladesh"
greeting = 'Hello, World!'
print(name)
print(greeting)

Explanation:
- Strings store words, sentences, or any text.
- You cannot perform mathematical operations on strings directly, but you can concatenate (+) or repeat (*) them.

Output:
Bangladesh
Hello, World!

5. Comparison Table

Data Type	Example	Description	Operations Allowed
Integer	10, -5, 0	Whole numbers without decimals	+, -, *, //, %
Float	3.14, -0.5	Numbers with decimals	+, -, *, /
String	"Hello"	Sequence of characters (text)	+ (concatenate), * (repeat)